Supplementary Material

Brain-Inspired 4-DoF Localization-Mapping: Complementary Vestibular-Visual Fusion with Spatial Cells

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S.1 Zero-Shot Failure of the Scale Head on KITTI 07

The monocular scale-correction head (Methods section) is trained once on CARLA imagery. On real KITTI 07 data its predictions invert: the head outputs $\log s \approx -1.1$ (i.e., $s \approx 0.3$, shrinking the already under-scaled VO), whereas the sequence requires a scale correction of $s \approx 1.8$. Deploying it — replaying the EKF with the head driving the scale state under the same deployment configuration as on CARLA (dead-band 0.30) — degrades the true-VIO EKF baseline from 17.75 m to 23.13 m (Sim(3) caliber): the filter's scale state collapses from its kinematically consistent value (≈ 5.5 in VO units) toward the erroneous deep prediction (≈ 0.5), dragging the fused trajectory with it.

Two further zero-shot reference points quantify what is recoverable without ground truth: the turn-kinematics scale observer (usable zero-shot by construction) converges to $s \approx 5.65$ (VO units) on this sequence, essentially the same value the EKF's scale state reaches on its own (≈ 5.54), while the true global scale is ≈ 6.78 — so the residual global-scale error is structurally out of reach for both zero-shot scale observers on this sequence. For this reason, all KITTI results in the main text run the loop-closure line over the unscaled EKF core, and the paper's metric-scale claim is carried by the four simulated scenarios plus the (scale-independent) KITTI loop-closure margin.

S.2 KITTI 07 Baseline Cross-Check (Dual Reporting)

The main-text KITTI 07 baseline is generated by our legacy degradation pipeline (visual odometry fused with IMU through the same 10-state EKF, with a scale-error profile representative of monocular deployment). Because that pipeline predates the final re-implementation, we additionally evaluate on a *true* visual-inertial baseline produced by the current public pipeline (`kitti_prepare_fusion.py`: ORB visual odometry fused with the OXTS IMU through the identical EKF):

KITTI 07 configuration	Sim(3) refit	SE(3) rigid
EKF (main-text baseline)	22.09 m	30.45 m
NeuroLocMap (loop line) on it	**18.45 m**	**29.99 m**
EKF (true-VIO pipeline, this section)	17.75 m	44.01 m
NeuroLocMap (loop line) on it	23.12 m	**40.99 m**

The comparison is instructive in both directions. Under the shape-only Sim(3) caliber the true-VIO baseline is stronger (17.75 m), and the loop line does not improve it (23.12 m): the Sim(3) refit already absorbs that trajectory's dominant error — a global scale factor of ≈ 1.8 — leaving a shape residual that the loop line's translation-only corrections cannot beat. Under the rigid SE(3) caliber, however, where the scale error *cannot* be absorbed, the loop line still improves its baseline (40.99 vs 44.01 m, -6.8%). This cross-check substantiates the paper's reading of the KITTI result: the experience-map loop closure provides genuine global-consistency margin on real data, and the

magnitude of that margin depends on how much global-scale error the caliber allows the aligner to absorb.

S.3 KITTI 09: The Persistence Gate as an Honest No-Op

To probe the gate chain beyond the five main-text scenarios we processed KITTI 09 (1591 frames, 1705.1 m, a large urban loop) with the identical zero-shot configuration. The result is a designed no-op: the loop line performs at parity with its EKF core (Sim(3) 42.37 vs 42.34 m; SE(3) 54.63 vs 54.77 m).

The trajectory contains exactly one genuine loop event — the final frames return to the start region (GT position gap 3.8 m). Both embedding front-ends rank this pair first with high confidence (zero-shot similarity 0.988, fine-tuned 0.994), but the revisit occurs in the last two frames of the sequence: the persistence gate (15 consecutive candidate frames, Methods section) can accumulate a streak of at most 2 before the sequence ends, and correctly declines to re-anchor. A what-if simulation that force-releases the event yields only -0.12 m (Sim(3)) — the loop arrives too late to matter. Meanwhile the gate chain rejects the two spurious high-similarity candidates (GT position gaps 90/134 m) via the orientation gate. The map itself is healthy (1491 experience nodes, 1474 visual templates).

A second, negative transfer result emerges from the embedding comparison: the fine-tuned embedding head — which improves precision on the CARLA towns — ranks the true loop pair *below* the acceptance gate on KITTI 09 (recall 0% at every threshold), while the zero-shot embeddings rank it first (recall 100%). Domain shift can silently degrade even a component that transfers on one real sequence (KITTI 07); we therefore report both variants and treat zero-shot embeddings as the default for cross-domain deployment.

S.4 Reproducibility Statement

All CARLA scenarios were collected with the public repository's collector (agent speed 12 km/h, 20 autopilot vehicles and 20 kinematically driven pedestrians) and are reproducible end-to-end from it. KITTI 07 and KITTI 09 are processed by the same public pipeline: raw sequences are assembled (`kitti09_assemble_raw.py`) and converted to the standard four-file format with a true visual-inertial EKF (`kitti_prepare_fusion.py`). Two baselines coexist for KITTI 07 and are reported side by side in Section S.2: the main-text baseline trajectory is preserved verbatim in the release (`_p11_backup_20261014/KITTI07/fp_old_synthetic.txt`), and the true-VIO baseline is regenerable from raw data by the current pipeline. All metric tables in this paper are computed from the released trajectory files by the evaluation scripts in the repository (`se3_metrics.py`, dual caliber), with cross-checks ensuring bit-consistency between tables.